

Literature Gap Tables

Privacy & Security in Distributed Machine Learning

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How to Read These Tables

Each table maps the **existing literature** on privacy and security attacks in a specific distributed machine learning paradigm. The goal is to make it immediately visible **what has been studied** (green/yellow cells), and **what remains unexplored** (red cells), guiding future research directions.

Tables (5 Paradigms)

1. **Federated Learning (FL)** — N clients train over multiple rounds with a central server aggregating model updates (FedAvg, FedSGD).
2. **One-Shot FL** — clients participate in only *one* round, typically via knowledge distillation (logit sharing).
3. **Split Learning (SL)** — the model is partitioned at a cut layer; intermediate activations (“smashed data”) are exchanged.
4. **Multi-Client SL / SFL** — combines SL with FL: parallel split training + federated aggregation.
5. **Multi-Server FL/SL** — distributes trust across multiple servers to eliminate single points of failure.

Columns (6 Attack Categories)

Abbr.	Category	What It Covers
P	Data/Model Poisoning	Label-flipping, clean-label, and gradient manipulation attacks that degrade accuracy.
B	Backdoor Attacks	Hidden triggers causing targeted misclassification while maintaining clean accuracy.
Byz	Byzantine Attacks	Worst-case adversarial deviations and robust aggregation methods (Krum, TrimMean).
R	Gradient/Data Reconstruction	Reconstructing private data from gradients, model updates, or smashed data.
I	Inference Attacks	Membership, property, attribute, label, and distribution inference.
S	Model Stealing	Model extraction, functionality stealing, and free-riding.

Rows (7 Adversary Models)

Row	Description
Malicious Client	Actively deviates from protocol; can send arbitrary updates.
Malicious Server	Tampers with aggregation, manipulates weights, or reconstructs data.
Mal. Collusion	Server + one or more clients conspire together.
Semi-Hon. Client	Follows protocol exactly but passively infers from observations.
Semi-Hon. Server	Follows protocol exactly but analyzes received updates.
Semi-Hon. Collusion	Multiple semi-honest parties share observations.
External Adv.	Eavesdrops on or tampers with communication channel.

Color Coding

Green (5+ papers) — well-studied. **Yellow (1–4 papers)** — limited coverage. **Red (0 papers)** — open research gap.

Table 1: Federated Learning (Standard Multi-Round)

Adversary	P	B	Byz	R	I	S
Mal. Client	[1, 2, 3, 4, 5, 6]	[8, 9, 7, 10, 11, 12, 13, 14]	[17, 18, 19, 15, 16, 20]	[21, 22, 23]	[27, 28, 29]	[33, 34, 35, 36, 37]
Mal. Server	[38, 39, 40]	[42, 41]	[41]	[43, 44, 45]	[46, 32]	—
Mal. Collusion	[47, 48, 49, 45]	[42]	[53, 54, 50, 51, 52]	[44, 45]	[55]	—
SH Client	[55]	—	—	—	[30, 31]	—
SH Server	—	—	—	[21, 22, 24, 25, 26]	[32]	—
SH Collusion	—	—	—	[44]	—	—
External	[56]	—	—	[57, 58, 59]	[60, 61]	[58]

Per-Cell Explanations:

Mal. Client $\times$ P:

[1] introduces label-flipping attacks under non-IID, showing heterogeneous data amplifies poisoning. [4] formalizes *local model poisoning* with optimization-based attacks against robust aggregators. [3] demonstrates multi-round consistency attacks that defeat 8 SOTA defenses. [2] proposes filtering poisoned updates via reconstruction error. [5] shows attackers can align perturbations with DP noise to achieve $\sim 100\%$ ASR while evading Krum/TrimMean. [6] demonstrates DP-exploited stealthy model poisoning hidden in noise distributions.

Mal. Client $\times$ B:

[7] introduces *constrain-and-scale* model replacement. [8] decomposes the trigger across clients for stealth. [9] targets infrequently-updated parameters for durability. [10] provides provable defenses reducing ASR to $< 8\%$. [11] detects backdoors via differential testing. [12] uses layer-wise scoring to capture full model information. [13] reduces ASR from 99.7% to 1.9% via neuron removal. [14] provides the first *certified robustness* framework for FL backdoors.

Mal. Client $\times$ Byz:

[15, 16] are foundational robust aggregation defenses. [17] optimizes poisoning against these specific defenses. [18] applies game-theoretic analysis to Byzantine defense. [20] hides the aggregation rule to shift attack impact to expected values.

Mal. Client $\times$ R:

[21] is the seminal gradient inversion attack. [22] improves it using cosine similarity matching. [23] recovers entire batches via batch normalization statistics.

Mal. Client $\times$ I:

[27] is the foundational membership inference paper. [28] combines poisoning with MIA for amplification. [29] uses poisoning to assist property inference.

Mal. Client $\times$ S:

[33] introduces free-rider attacks where clients steal models without contributing data. [34] detects free-riders via MIA/property inference. [35] uses weight evolving frequency analysis for early detection. [36] applies deep autoencoding Gaussian mixture models. [37] studies model IP cloning from eavesdropped FL traffic.

Mal. Server $\times$ P:

[38] proposes a two-server FHE defense with anomaly detection. [39] shows adversarial weight initialization facilitates data extraction. [40] uses adaptive DP for non-fully trusted servers.

Mal. Server $\times$ R:

[43] reconstructs local samples even under secure aggregation. [44] provides a comprehensive taxonomy of server-side GIAs. [45] breaks SecAgg by modifying a single BN layer, achieving 100% label recovery.

Mal. Collusion $\times$ Byz:

[50] shows sybils can prevent convergence entirely. [51] defends via cosine similarity of historical updates. [52] uses gradient diversity to detect coordinated sybil attackers.

External $\times$ R:

[57] analytically reconstructs client local models from wiretapped updates. [58] proves full-model uploads leak more IP than delta uploads.

2 Table 2: One-Shot Federated Learning

Adversary	P	B	Byz	R	I	S
Mal. Client	[63, 62, 64]	[10, 11, 76, 77]	[78, 79, 80, 81, 82, 83, 84]	—	—	—
Mal. Server	[65, 66, 67]	—	—	[68, 69, 70]	[71, 70]	—
Mal. Collusion	—	—	—	—	—	—
SH Client	—	—	—	—	—	—
SH Server	—	—	—	[73, 68]	[72, 71]	—
SH Collusion	—	—	—	—	—	—
External	—	—	—	—	—	—

DP+Poisoning: [5] hides in DP noise ($\sim 100\%$ ASR). [6] exploits DP distributions. [74] decouples via sign aggregation. [75] uses Fourier filtering.

Per-Cell Explanations:

Mal. Client $\times$ P:

[62] is the first logit poisoning attack in FD: sorts and remaps confidence values to mislead aggregation. [63] controls peak logit values for stealthier manipulation. [64] uses K-means clustering to shuffle logit groups and amplifies with scaling.

Mal. Client $\times$ Byz:

[79] combines hybrid-domain fuzzy clustering with meta-learning for one-shot Byzantine detection. [80] aggregates predictive posteriors instead of model parameters, outperforming multi-round in non-IID. [81] treats aggregation weights as learnable parameters with convergence guarantees. [82] uses loss-based greedy selection on server reference data, designed for non-IID. [83] employs credibility-based adaptive anomaly detection. [84] simulates heterogeneity impact during aggregation. [78] provides robust aggregation against top-k and impersonation attacks.

Mal. Server $\times$ P:

[65] shows a malicious server can curate the public dataset to steer the model. [66] demonstrates server tampering with aggregated logits before redistribution. [67] defends via outlier scoring across the public dataset.

Mal. Server $\times$ R:

[68] introduces Paired-Logits Inversion, recovering images from logit pairs in FedMD. [69] extracts labels from uploaded logits. [70] crafts public samples to infer label distributions and membership from client logits.

Mal. Server $\times$ I:

[71] (our work) shows LDP noise does not reduce MIA AUC (0.59–0.70); HE required. [70] enables server label distribution inference via carefully crafted transfer set queries.

SH Server $\times$ I:

[72] audits privacy loss in a single run (ICLR’24). [71] finds HE is more effective than LDP alone for MIA defense.

3 Table 3: Split Learning

Adversary	P	B	Byz	R	I	S
Mal. Client	[85, 86]	[88, 89, 90]	—	—	[114, 115]	[117, 118, 119, 120]
Mal. Server	[85, 87]	[91, 92, 93, 94, 85]	—	[95, 96, 97, 98, 99, 100, 101]	[105, 106, 107, 108, 109, 110, 111]	[95]
Mal. Collusion	[104, 103]	—	—	[102, 103]	—	—
SH Client	—	—	—	[116]	—	—
SH Server	—	—	—	[125, 126, 121, 122, 123, 124]	[112, 113]	—
SH Collusion	[127]	—	—	[121]	—	—
External	[130]	—	—	[121, 128, 129]	—	—

Per-Cell Explanations:

Mal. Client $\times$ S:

[117] proposes 5 gradient-query variants achieving >90% extraction accuracy of the server-side model. [118] uses a surrogate bottom model with generators to steal client-side embeddings. [119] demonstrates data-oblivious model inversion and stealing. [120] proposes defenses satisfying transparency, effectiveness, and scalability.

Mal. Server $\times$ B:

[91] backdoors vertical SL with >90% ASR at 1% poison rate using embedding-space triggers, despite having no label access. [92] achieves VFL backdoor via label inference + poison phase. [93] defends by purifying backdoor-triggered embeddings at inference. [94] shows feature-based attacks succeed without explicit triggers.

Mal. Server $\times$ R:

[95] enables pseudo-client data stealing without knowing the client architecture. [96] introduces stealthy feature-oriented reconstruction (1.97 $\times$ better than FSHA, semi-honest). [97] hijacks the feature space to reconstruct client data (active attack). [100] defends using function secret sharing (>96% accuracy preserved). [101] detects hijacking by monitoring gradient pattern anomalies. [98] exploits representation preferences in smashed data.

Mal. Server $\times$ I:

[108] achieves F1 > 0.99 on label reconstruction via clustering of gradients and activations. [109] uses cosine/Euclidean similarity matching with one auxiliary sample per class. [110] frames label leakage as supervised learning, reaching 97% accuracy. [105] uses adversarial regularization for passive MIA from smashed data. [111] defends via bidirectional mapping to increased K-class labels.

SH Server $\times$ R:

[121] encrypts only server-side parameters via CKKS HE with 16 $\times$ efficiency gain. [122] achieves 2.5–10 $\times$ faster inference with 14–290 $\times$ communication reduction. [123] allows server computation on encrypted activations with only 2.65% accuracy loss. [124] reduces overhead by 90% by encrypting only critical parameters. [125] transfers resistance properties against model inversion.

SH Server $\times$ I:

[112] randomizes ReLU activations to defend against property inference and reconstruction. [113] encrypts the exchange tunnel to limit server correlation.

Mal. Collusion $\times$ R:

[102] combines client smashed data + server gradients + output layer for label recovery. [103] shows adjacent nodes in multi-hop chains can reconstruct data by combining partial views.

4 Table 4: Multi-Client Split Learning / SFL

Adversary	P	B	Byz	R	I	S
Mal. Client	[132, 133, 134]	[90, 88]	—	—	[114]	[117, 131, 119]
Mal. Server	[38, 87]	[85, 91]	—	[95, 96, 97, 99, 98, 121]	[105, 107, 108]	[95]
Mal. Collusion	[104]	—	—	[102, 103]	—	—
SH Client	—	—	—	[115]	—	—
SH Server	—	—	—	[125, 121, 124]	[135]	—
SH Collusion	—	—	—	—	—	—
External	—	—	—	[136]	—	—

Per-Cell Explanations:

Mal. Client $\times$ *P*:

[132] studies data poisoning attacks (DPA) on ECG/MNIST under SFL. [133] proposes a unified defense for SFL poisoning. [134] shows SFL inherently reduces accuracy degradation by $5\times$ vs FL due to its architecture.

Mal. Client $\times$ *S*:

[117] applies the 5 gradient-query extraction variants in multi-client settings. [131] protects server-side model IP via partitioned access control. [119] is applicable for data-oblivious model inversion in multi-client SL.

Mal. Server $\times$ *R*:

All single-SL server attacks ([95, 96, 97, 99, 98]) are applicable; multiple client representations may provide richer reconstruction signals. [121] serves as defense using HE for encrypted server-side computation.

5 Table 5: Multi-Server FL/SL

Adversary	P	B	Byz	R	I	S
Mal. Client	[137, 138]	—	—	—	—	—
Mal. Server	[38, 137]	—	—	[139]	—	—
Mal. Collusion	—	—	—	—	—	—
SH Client	—	—	—	—	—	—
SH Server	—	—	—	[140, 141]	—	—
SH Collusion	—	—	—	—	—	—
External	—	—	—	—	—	—

Per-Cell Explanations:

Mal. Client $\times$ P:

[137] (TU Delft thesis) shows multi-server FL dilutes client poisoning impact; however, all tested defenses proved ineffective against FMes/Min-Max attacks.

Mal. Server $\times$ R:

[139] uses lightweight MPC so neither server alone accesses plaintext gradients. [38] applies a two-server defense with fully homomorphic encryption.

Mal. Collusion (all cells absent):

This is a **critical gap**: multi-server designs distribute trust, but what happens when servers collude to combine partial views is completely unexplored.

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